Imran E Kibria

4^{th} YEAR PHD RESEARCHER IN SPEECH PROCESSING

kibria.5@osu.edu | Portfolio | Github | Scholar | LinkedIn

Research Interests

Speech Quality Assessment, Speech Quality Enhancement, Auditory Attention Detection, Audio Explainability.

Experience

Graduate Teaching Associate — *Columbus, OH, USA*
Department of Computer Science & Engineering, Ohio State University — *Aug 2023 - Present*

Courses

- Machine Learning and Statistical Pattern Recognition (Spring 2026)
- Software I: Software Components — Course Instructor (Summer & Fall 2025)
- Introduction to Neural Networks (Fall 2024)
- Introduction to Computer Programming in Python (Fall 2023; Spring 2024, 2025)

Graduate Research Associate — *Columbus, OH, USA*
Audio, Speech, & Perceptually-Inspired Research Lab, Ohio State University — *May 2023 - Aug 2023*

Design Engineer — *Islamabad, Pakistan*
Center for Advanced Research in Engineering — *Nov 2020 - May 2022*

Developer Intern — *Islamabad, Pakistan*
N-ovative Health Technologies — *July 2020 - Oct 2022*

Education

Ohio State University — *Columbus, OH, USA*
Ph.D. in Computer Science & Engineering, GPA: 3.9 / 4.0 — *Aug 2023 - Present*

- Research: Neural Representations for Perceptual Speech Quality
- Advisor: Donald S. Williamson

National University of Sciences and Technology — *Islamabad, Pakistan*
BS in Electrical Engineering — *Sept 2016 - July 2020*

- Major: Signal Processing and Machine Learning
- Thesis: Real-time Adaptive Filtering for Robust Vibration Sensing

Projects

Lightweight AI for Perceptual Speech Quality on the Edge [Code]

- Developed an attention-based MOS prediction model with only 87K parameters.
- Investigated transformer design choices, including the role of positional encodings.
- Achieved performance comparable to wav2vec 2.0 while using 100× fewer parameters.
- Improved noisy-label robustness through self-teaching, increasing correlation by 8%.

Skills: Swin Transformers, Knowledge Distillation, Foundation Models, Python, TensorFlow, SHEET Toolkit.

One Model for All Domains: Zero-Shot Speech Quality Assessment [Code]

- Studied domain shift and range-equalizing bias across seven speech quality datasets.
- Applied Sharpness-Aware Minimization to improve cross-domain generalization.
- Analyzed loss landscapes and model flatness to explain robustness gains.
- Reduced MSE by up to 76% and improved correlation by 42% across eleven test sets.

Skills: Domain Generalization, Sharpness-Aware Minimization, Python, TensorFlow, SHEET toolkit.

Self-Supervised Learning for Speech Quality Assessment [Code]

- Adapted AST, CLAP, HuBERT, wav2vec 2.0, and WavLM for MOS prediction.
- Evaluated transfer learning performance on the NISQA benchmark.
- Compared SSL representations for robustness and cross-dataset generalization.

Skills: Self-Supervised Learning, Transfer Learning, Domain Generalization, Python, PyTorch, HuggingFace.

Smartphone-Based Urine Test Strip Analysis for At-Home Diagnostics [Code]

- Developed a smartphone-based system for automated urine test strip analysis.
- Combined color calibration via linear regression with Euclidean color matching.
- Achieved 92% in-house accuracy as a low-cost alternative to laboratory analyzers.

Skills: Urinalysis, Image Segmentation, Chromatic Calibration, Color Spaces, Python, Scikit-learn.

Technologies

Languages: Python, MATLAB, Java, C, C++, SQL, LaTex.

Deep Learning: Tensorflow, PyTorch, HuggingFace, Github

Speech Toolkits: SHEET, ESPNet, SpeechBrain

Awards

CCBS Summer Graduate Research Award June 2026
Grant approved by The Center for Cognitive and Brain Sciences (CCBS) to study human speech perception

CSE Scarlet & Gray Award Aug 2023
Merit-Based Scholarship, The Ohio State University (5-year award)

Best English Speaker June 2017
SEECS (School of Electrical Engineering & Computer Science) Declamation Contest

Certificate of Excellence Dec 2014
Third Place – Secondary Board Examination, Cadet College Hasanabdal

Publications

A Generalization Strategy for Speech Quality Prediction: Domain-Specific to Unified Datasets May 2026
Imran E Kibria, Ada Lamba, Donald S. Williamson
IEEE ICASSP 2026

AttentiveMOS: A Lightweight Attention-Only Model for Speech Quality Prediction Aug 2025
Imran E Kibria, Donald S. Williamson
ISCA Interspeech 2025

Smartphone-Based Point-of-Care Urinalysis Assessment July 2022
Imran E Kibria, Hussnain Ali, Shoab A. Khan
IEEE EMBC 2022